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Screenless Completion Applications Restore Sand-Free Production with Cost Effective Intervention in La Yuca Field, Colombia

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Abstract

Sand production is a common problem for oil and gas wells producing from the younger age unconsolidated clastic reservoirs. Formation solids production will cause operational problems varying from expensive sand handling problems to the complete loss of a productive zone or even the possibility of lost well control, due to eroded surface or even abandonment. This is the case of La Yuca field in Colombia, where sand production is a common problem leading to frequent artificial lift pump failures, lost production and well shutdowns.

Several methods exist to limit formation sand and fines production, from sand and production management to changes in the completion architecture, including perforation strategies, standalone screens, open and cased hole gravel and frac-packing techniques. These last two include the use of a screen to hold the control gravel or proppant in place acting as the retention media to the formation sand.

A method known as screenless completions advocates the elimination of the screen to hold proppant in place. Advantages of such technique are a simpler method and intervention inherent to the elimination of screens and associated jewelry, lower cost and maintaining full wellbore access for future interventions. This is in addition to gaining sand control and providing stimulation through fracturing.

However, when the application involves high rates, low temperature and limited closure stress it has been historically difficult to keep proppant in place. A process of locking the proppant has been developed using a two-part polymeric system that works in a wide range of temperature and stress conditions. This two-part system consists of a proppant with a coating and a chemical activator that react to create bonds to create a self-supporting proppant pack. This system has been deployed in several wells in La Yuca effectively restoring wells to sand-free production while simultaneously stimulating the production capacity of the wells.

This paper will present the basis of the technology and two screenless completion application case histories in Colombia. It will be helpful to engineers looking to control formation sand giving guidance in candidate selection and screenless completion design in a variety of remedial applications.

Introduction

The Llanos Basin area in Colombia is located in a vast, flat grassland region sloping gently eastward from the foot of the Eastern Cordillera (Figure 1). From April to November heavy rains soak the region, which is typically very dry the rest of the year. Some areas are swamps, even during most of the dry season. The surface is entirely covered by recent deposits, such that both reflection seismic and geochemical studies, which corroborate and support the initial seismic interpretation, are the only effective exploration methods prior to drilling.

Figure 1—Llanos Basin location in Colombia

The Cravo Norte Association concession is located in the northern portion of the Llanos basin. Overall, the field is a large ellipsoidal feature influenced by the Arauca Arch, but can be further divided into several separate fault blocks controlled by major strike slip faults, each with separate oil-water contacts. The structure is controlled by three main strike slip faults: the Caño Limón, La Yuca and Matanegra faults that separate the area into distinct structures. The La Yuca strike slip fault has a maximum vertical offset of about 400 feet and separates the La Yuca Este downthrown block from La Yuca upthrown block.

La Yuca field

The La Yuca field stratigraphy in the Northern Llanos basin is characterized by Tertiary and Cretaceous formations, particularly the Upper Carbonera (C1 to C5), Lower Carbonera (M1 to M3), and Cretaceous K1 and K2 oil producing reservoirs (Ellison and Ahumada 1995).

The reservoirs are Cretaceous to Oligocene aged sandstones deposited in an overall regressive system, grading from shallow marine, through deltaic to fluvial depositional environments with an average formation depth of 7,450 ft, average porosity of 28%, average permeability of 50 mD–6,000 mD, and recovery factors that range between 30% to 70%. The formations are poorly consolidated and tend to produce sand to varying degrees (Ellison 1995).

Reservoir fluid properties are somewhat unusual. Although the API gravity is high, the gas-oil ratio and the bubble point pressure are very low. The recovery mechanism is a strong edge-water drive, which leads to high water cuts that can exceed 90%. As a result of the combination of modest pressure depletion, low GOR and high water cut, wells won't naturally flow to surface and must be produced using ESP's, PCPs or rod pumps.

The Sand Production Challenge

Production rates and drawdown are limited to prevent sand production. In this manner, formation solids are kept under control to protect artificial lift equipment from erosion damage. This strategy is quite successful, particularly during initial production. However, at some point in time most wells present sand production related issues, most commonly fill in the wellbore and lift pump failures, leading to well shut-in and deferred production. These occurrences demand rig intervention to pull the completion, clean up the fill and re-run a new pump.

Key performance indicators. Two key indicators to track well performance from a solids control perspective are *sand fill*, expressed in ft/month, and *run time* defined as time elapsed between artificial pump failures, expressed in days. *Run Time* specific KPI target is 1,000 days.

In terms of economic indicators, direct cost of operations involving workover remedial operations and equipment replacement costs are at the top of the list. In addition to direct costs, shut-in times and deferred production are also primary considerations due to the difficult access to the area due to geography, infrastructure, security and community restrictions that may delay repair work for weeks or months.

Wellbore Access. Sand production leads to perforation and wellbore destabilization as well as ID restrictions, causing limitations to wellbore access to run mechanical and logging tools, screens, and completion accessories.

New approach

The proposed solution for La Yuca field was to place a skin by-pass screenless completion to control formation solids, comprised both of sand and fines ingress, prevent proppant flowback and stimulate the formation simultaneously. Given the proximity of water bearing zones below the treated interval, the frac had to be designed such that it would by-pass the near wellbore damaged zone without risking breaking into the water. The details of the technology will be presented in the following sections and then illustrated with case histories from the field.

Self-Supporting Proppant

A novel Self-Supporting Proppant (SSP) technology was selected to place the skin by-pass fracpacks. Rather than relying on a sand screen to hold the proppant in place, the proppant itself is self-supported by forming a highly permeable consolidated pack. Besides providing stimulation and sand control, this approach enables maintaining full wellbore access and eliminates the need to run costly screen completions and associated jewelry.

The candidate wells had been producing sand for some time, consequently, to achieve effective sand control, the proppant not only has to consolidate in the frac, but also in potential voids behind casing and in the existing perforations. These are challenging conditions given voids and perforations are areas of no stress. Under these conditions grain-to-grain contact is very loose, therefore compromising the ability of conventional resin-coated proppants to bond despite curing of the resin.

The selected SSP technology consolidates without the need to be under confinement stress, as well as at temperatures from ambient to 600°F. The system has two parts, a specialized resin coating applied at the manufacturing plant, and a companion liquid activator. Both components are stable by themselves at storage conditions, with bonding only occurring following a chemical reaction between coating and activator once mixed in the field (Whaley 2016, Leasure 2014).

In operations, the activator is first added to the fluid at the suction of the blender (for continuous operations) or batch mixer tank (for batch mixed operations). Since the activator is highly hydrophobic, it is dispersed in small droplets into water-based carriers, rather than going in solution. When the proppant is added, the oil wet nature of its surface creates a strong affinity for the hydrophobic activator to immediately adsorb onto the resin coating to begin the activation sequence. The activated proppant and activator system is then pumped into place to complete the reaction, bonding and consolidation of the pack. The technology also incorporates a window of working time to allow the activated proppant to travel to and into the fracture, as well as time to reverse out or clean out the well should a premature screen-out occur. The working time is dictated primarily by the bottom hole temperature at the end of pumping and for typical application temperatures is in the range of several hours. Once the working time has elapsed it takes several hours to days (depending on bottomhole static temperature (BHST)) for the SSP to reach the ultimate strength, measured by its Unconfined Compressive Strength (UCS). The final UCS is primarily determined by BHST, stress on proppant, and carrier fluid and can be further controlled by the SSP activator concentration.

Field implementation

Two wells were selected for the initial implementation and proof of concept in La Yuca field. Both wells suffered from sand production that affected the pumps installed for artificial lift, leading to failure and frequent changes. Given these wells had existing perforations and may have voids behind casing due to sand production, the SSP was selected to ensure consolidation in these no-stress zones. In this manner all paths connecting the formation with the wellbore would be protected with an in-situ formed filter to prevent sand influx.

Even though a skin bypass frac was desired its extension should be limited to avoid breaking into the water zones. The technique selected for this purpose was a High Rate Water Pack (HRWP) or extension pack. Due to the high permeability and unconsolidated nature of the formation, it is difficult to reach a frac regime, therefore the jobs were designed to be pumped at high rates with a low efficiency fluid so that the proppant would successfully fill caverns away from the wellbore if a frac is not attained.

Case History 1 – La Yuca 232

La Yuca 232 well was drilled in 2011 as a single zone vertical producer from the K1B1 unit. It was completed with 9 5/8″ cemented casing, perforated in the 7,510–7,518 ft interval and produced with an ESP artificial lift pump. Despite being produced at a controlled draw down of ~300 psi, the well presented several pump failures that led to long shut-in periods between 2012 and 2023 when it was selected for a remedial treatment with SSP. The ESP record shows 6 failure events, with 4 of them caused by sand production (Table 1). The run time between sand-caused failures is 12 months on average and 7 months for the shortest. The average shut-in period following sand-caused failures is 7 months on average and 16 months for the longest.

Table 1—Sand Rate Determination. Wireline, Slickline and Drill Pipe interventions were conducted as part of the sand management strategy to determine sand fill, and not due to ESP nor well productivity issues.

Well Intervation Date	Intervention	Well Bottom ft	Method Tag Bottom	Rathole, ft	Sand fill ft	Run Time [Days]	Sand Rate ft/month	Sand Clean Out Required?	Tag Depth ft-MD.
19-Oct-11	Initial Completion	7,841	Wire Line	323	-	-	-	-	-
25-Mar-12	2	7,528	Slick Line	10	313	123	77.9	Yes	7,841
12-Jan-13	3	7,537	Slick Line	19	304	215	43.1	Yes	7,826
21-Dec-17	4	7,516	Drill Pipe	-2	310	445	21.2	Yes	7,790
25-Mar-19	5	7,515	Wireline	-3	275	444	18.9	Yes	7,841
21-Nov-20	6	7,512	Wireline	-6	329	307	32.7	Yes	7,841

The last reported production data from the well on December 2020 prior to ESP pump failure indicated a gross rate of 3,700 BFPD with a water cut above 90% (Figure 3). The mechanical state of the well at this time showed fill in the perforation interval (Figure 2).

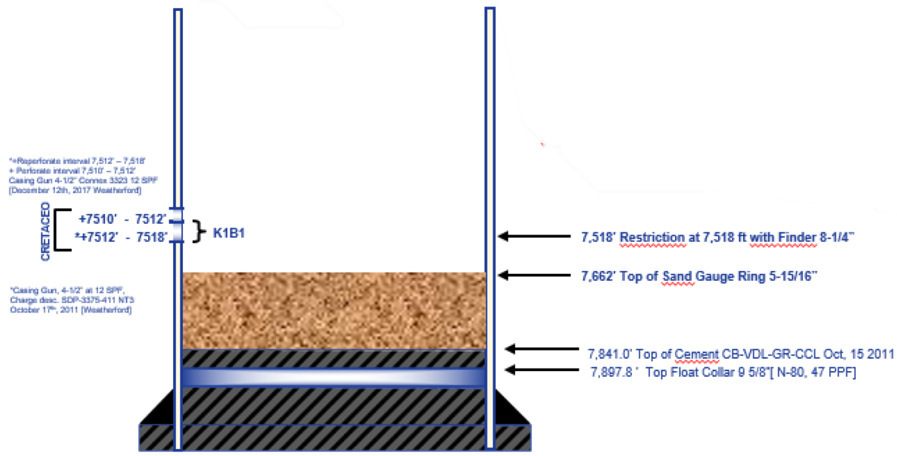

Figure 2—LY-232 Mechanical Status

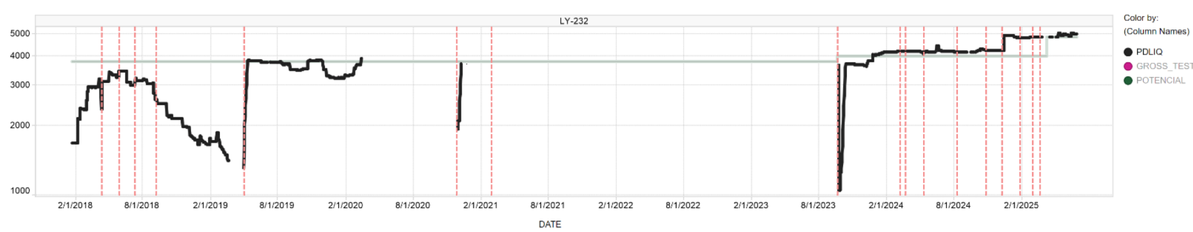

Figure 3—LY232 Fluid Production History

Screenless skin by-pass design. The objective of the intervention was to recover the well potential to last reported production, and remain sand free for an extended time without further ESP breakdowns. The high-level design comprised the following steps:

- Pull ESP, clean out to TD, run work string & packer
- Perform injectivity test with brine: determine frac extension rate (FER)
- Place HRWP with 1% NH_4Cl brine 2 BPM above FER
  - 6,400 lb planned 0.5 – 1.0 PPA
  - Pad 2,100 gal
  - P_{net} target 1,000 psi (from treating pressure)
- Shut-in well for 24 hr. to allow SSP to bond and consolidate
- Cleanup under flushed proppant if set across perforations (optional)
- Run ESP and produce well

The treatment was designed as a short skin by-pass HRWP targeting 30 ft of penetration, 3 lb/ft^2 proppant concentration with 1,000 psi Net Pressure (P_{net}) gain in a Tip Screen Out (TSO) design. The primary properties of the target formation used in the design are listed below:

- Reservoir Pressure: 1,400 psi (0.18 psi/ft)
- Permeability: 600 – 1,000 mD
- Porosity: 25%

- BHST 183°F
- Closure Stress ~3,600 psi (FG = 0.48 psi/ft)
- Young's Modulus: 1.43 – 1.99 E6psi, UCS = 368 – 620 psi

There were two data points for the formation sand Particle Size Distribution (PSD) – one from sand bailer (June 2017) and the other a de-sander pump run (December 2017) - indicating a mean diameter D-50 of 325 µm and 270 µm respectively (Figure 4). Based on these D-50s a 12/18 mesh proppant would be adequate to control the formation sand applying Saucier's criteria. However, based on a combination of field history and the possibility of bailed samples being segregated and coarser than formation sand, a more conservative approach was taken, with a 20/40 mesh proppant being utilized.

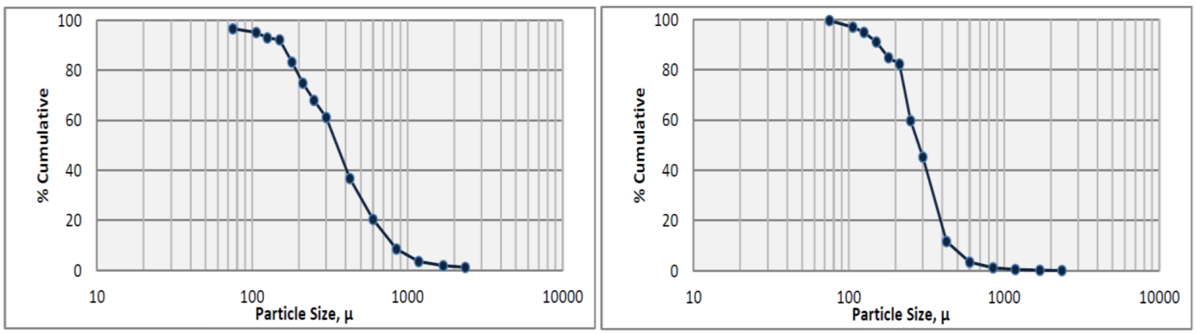

Figure 4—Formation Particle Size Distribution recovered from LY-232 from sand bailer (left) and de-sander pump (right) runs

Job execution. The job was pumped in September 2023, starting with a breakdown, Step Rate Test (SRT) and Step Down Test (SDT) to determine the FER. The FER was determined at 12 BPM and the pump rate for the main job set at 14 BPM, 2 BPM above FER.

The HRWP was pumped utilizing 378 BBL of NH_4Cl brine and 8,900 lb of SSP 20/40 at 0.5 – 1.0 PPA, extending it from the planned 6,400 lb. The average pressure during the job was 2,550 psi, reaching a maximum of 4,000 psi at shutdown. Good packing of the frac was achieved with a P_{net} gain of 1,600 psi (Figure 5). Displacement was completed to the top of perforations, leaving a minimum amount of under flushed proppant in casing. Enough rathole was available in the well (277 ft) for proppant to settle, and consequently a post job cleanout was not needed.

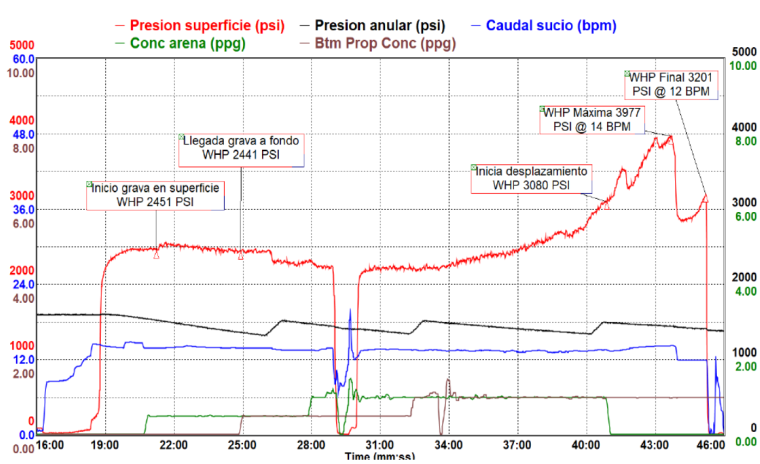

Figure 5—LY-232 job execution chart

Results. The well was shut-in for 24 hr. following the HRWP job, a cleanup post job was not required given the rathole was large enough to hold small amount of under flushed proppant A new ESP was run and the well was opened to production on September 23rd, 2023. Initial production reported 3,700 BFPD gross production, at a controlled drawdown of 500 psi with a PI of 4.12 BBL/psi. The well is currently being managed at a drawdown of 1000 psi without sand production, allowing to produce more fluid.

Most importantly, the well has been operating trouble free for +22 months with no sand or proppant flowback production as of this writing, preventing ESP failure for over a year and a half. This represents an 83% increase in the average ESP run time to date.

Note that a reduction in PI was observed following the intervention (Table 2). This is due to that, after filling the voids behind casing, the well is flowing through a proppant pack with an associated delta-P, rather than freely through empty space with zero pressure drop. Nevertheless, recovering the well to sand-free production with an extended pump run time, has been worth this compromise.

Table 2—LY-232 results after SSP

Intervention	BFPD	BPI (BOPD/psi)	PI (BFPD/psi)	Drawdown (psi)	RunTime (days)	Sand (ppm)
Before SSP	3695	1.87	8.22	450	18	20
After SSP	4995	0.93	4.12	1012	660	0

Case History 2 – La Yuca 276

La Yuca 276 well was drilled in 2015, cased with a 7″ cemented casing and completed as a single zone producer from the K1A2 unit. In 2022 it was recompleted to the C5c unit by isolating the lower perforations with bridge and cement plugs, and perforating the 7,359 ft – 7,379 ft interval (Figure 6). An artificial lift beam pump was installed to produce the well (Figure 7).

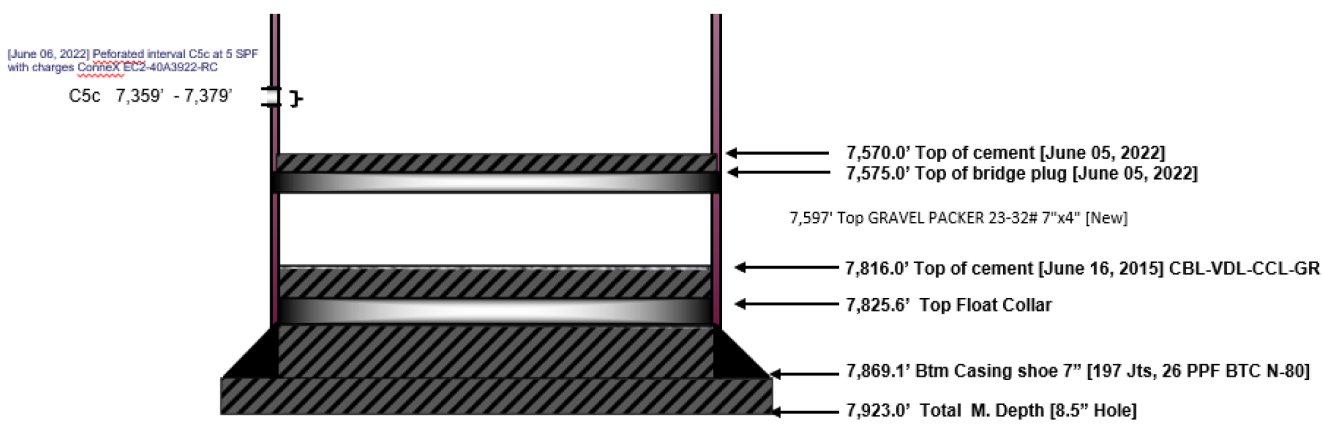

Figure 6—LY-276 Mechanical Status

Figure 7—LY-276 Fluid Production History

The beam pump record shows 3 failure events, with 2 of them caused by sand production (Table 3). The run time between sand-caused failures is just 1 ½ months on average and only 3 weeks for the shortest. The average shut-in period following sand-caused failures is 5 months on average and 9 months for the longest, occurring right before planning the SSP remedial job.

Table 3—Wireline, Slickline and Drill Pipe interventions were conducted as part of the sand management strategy to determine sand fill, and not due to ESP nor well productivity issues.

Well Intervention Date	Intervention	Well Bottom ft	Method Tag Bottom	Rathole, ft	Sand fill ft	Run Time Days	Sand Rate ft/month	Sand Clean Out Required?	Tag depth ft-MD
2-Jun-22	Initial Completion	7,570	Wire Line	191	-	-	-	-	-
4-Nov-22	1	7,325	Slick Line	54	137	55	74	Yes	7,570
13-Sep-23	2	7,485	Slick Line	106	81	22	110	Yes	7,570

The objective of the intervention was to recover the well potential to last reported production and remain sand free for extended time without further artificial lift pump breakdowns. The high-level program comprised the same steps as LY-232 as follows:

- Pull beam pump, clean out to TD, run work string & packer
- Perform injectivity test with brine: determine frac extension rate (FER)
- Place HRWP with 1% NH_4Cl brine 2 BPM above FER
  - 6,400 lb planned 0.5 – 1.0 PPA
  - Pad 2,100 gal
  - P_{net} target 1,000 psi (from treating pressure)
- Shut-in well for 24 hr. for SSP to bond and consolidate
- Cleanup under flushed proppant if set across perforations (optional)
- Run beam pump and produce well

Screenless skin bypass design. The treatment was designed as a short skin by-pass HRWP targeting 30 ft of penetration, 3 lb/ft^2 proppant concentration with 1,000 psi Net Pressure (P_{net}) gain in a Tip Screen Out (TSO) design. The primary properties of the target formation used in the design are listed below:

- Reservoir Pressure: 1,000 psi (0.14 psi/ft)
- Permeability: 160 mD
- Porosity: 21%
- BHST 191°F
- Closure Stress ~3,300 psi (FG = 0.45 psi/ft)

No formation sand PSD was available in the job design stage, however a bailed sample was collected in September 2023, just prior to the job (Figure 8). The PSD indicated a mean diameter D-50 of 220 µm. Applying Saucier's criteria a 16/20 mesh or finer proppant was found to be adequate to control the formation sand. Based on availability and to coordinate logistics with LY-232 operations, the same 20/40 mesh proppant size was proposed for LY-276.

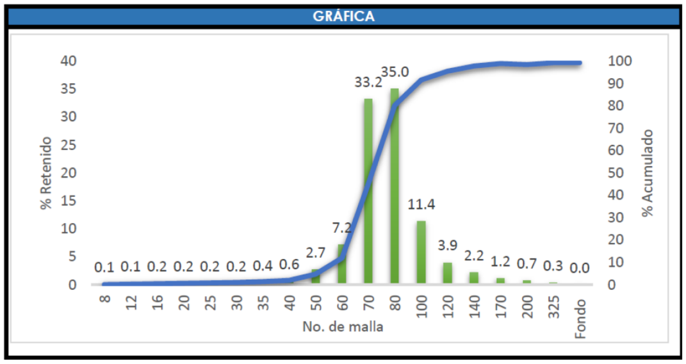

Figure 8—Formation Particle Size Distribution recovered from LY-276 from sand bailer.

Job execution. The job was pumped in September 2023, starting with a breakdown, SRT and SDT to determine the FER. The injectivity in this first phase was deemed low with a FER of 12.6 BPM, consequently a second SRT was planned to verify results. The second SRT showed better injectivity and a lower FER of 7.8 BPM. The fluid efficiency from these tests was 50%, high for the brine used, and the closure stress 3,300 psi (BH), in line with expectations.

The HRWP was pumped at 13 BPM utilizing 411 BBL of 1% NH_4Cl brine and 7,700 lb of SSP 20/40 at 0.5 – 1.0 PPA, extending it from the planned 6,400 lb. The average pressure during the job was 4,900 psi. The pump rate was reduced during the displacement to try and induce packing of the frac achieving a modest 160 psi P_{net} gain comparing the ISIPs of SRT (1,640 psi) vs main job (1,800 psi). Displacement was completed to the top of perforations, leaving a minimum amount of under flushed proppant in casing. Enough rathole was available in the well (191 ft) for proppant to settle, consequently a post job cleanout was not needed (Figure 9).

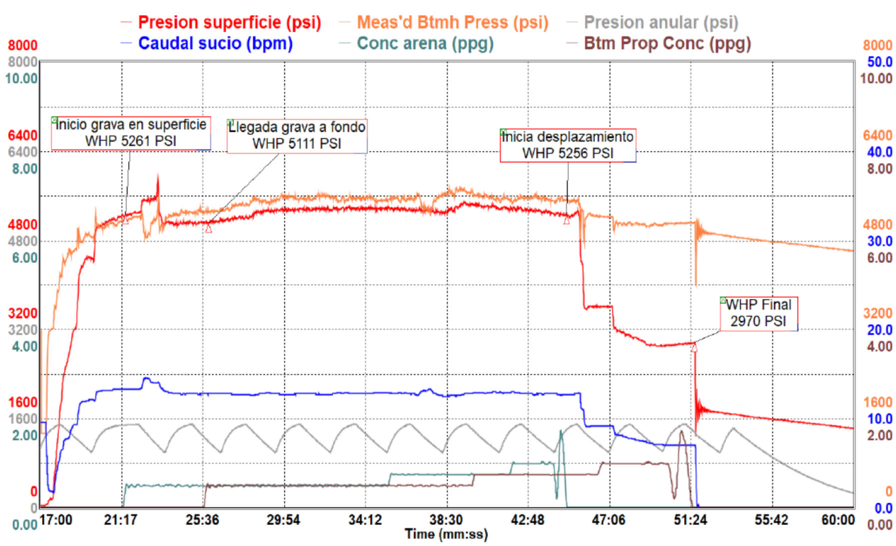

Figure 9—LY-276 job execution chart

Results. The well was shut-in for 24 hr. following the HRWP job, after which a new beam pump was run and the well was opened to production on October 16, 2023. A cleanup post job was not required given the rathole was large enough to hold small amount of under flushed proppant. In November 2024 the pump rod parted and was replaced, which was not associated to sand production. That aside, well LY-276 has been operating trouble free for +22 months with no sand or proppant flowback production extending the run time of the artificial pump for time over a year and a half at the time of this writing (Table 4). This extended run time represents a 14-fold increase on the average run time of 1 ½ months prior to the intervention.

Table 4—LY276 results after SSP

Intervention	BFPD	BPI (BOPD/psi)	PI (BFPD/psi)	Drawdown (psi)	RunTime (days)	Sand (ppm)
Before SSP	156	0.35	0.46	339	21	0
After SSP	120	0.175	0.34	353	660	0

Similar to LY-232, a reduction in PI was observed following the intervention. Once again, after filling the voids behind casing, the well is flowing through a proppant pack with an associated delta-P, rather than freely through empty space with zero pressure drop. Nevertheless, recovering the well to sand-free production with an extended pump run time has more than made up for the lower PI.

Summary

Sand production is a persistent issue in La Yuca field in the Llanos basin in Colombia. Controlled drawdown and production rate is applied with reasonable success to reduce sand influx, however this methodology has failed in more demanding cases. A new solution has been implemented based on placing a skin by-pass screenless fracpack to control formation solids, with the aid of an Self-Supporting Proppant system to keep the pack in place. The technique has proven successful in the two wells where it was initially trialed by extending artificial pump run times and preventing long shut-in times due to failures.

At the time of writing the pump run time in LY-232 and LY-276 wells has reached 22 months and is still going. This represents a +2.6-fold increase compared to the 8 ½ months historical average for both wells. Additionally, there have not been indications of proppant presence in the production facilities nor the wells' rat hole.

By preventing pump failures, the long shut-in periods derived from delayed intervention are also prevented. Considering shut-ins post failure have averaged 6 months, the benefit of avoiding them has been very significant in terms of saving deferred production.

A reduction in PI was observed following the intervention in both wells. This is because after the intervention the wells are flowing through a proppant pack with an associated delta-P, rather than freely with no pressure drop through voids derived from the sand produced. The benefits of recovering the well to sand-free production with an extended pump run time more than offset the impact of this reduction.

Additional benefits of the newly implemented technique are maintaining full wellbore access and completion diameter, as well as cost savings associated with not requiring running screens and associated jewelry. The SSP holds itself in place creating an in-situ filter in the frac, potential voids behind casing and existing perforations for effective sand control.

This work has proven the applicability of the screenless fracpack completions concept to La Yuca field in the Llanos basin as a valid technique to control sand production and mitigate failure of artificial lift equipment. This opens new opportunities to apply the technology in the basin, as well as other areas with similar conditions.

Glossary

BBP Barrels
BHST Bottom Hole Temperature
BPM Barrels Per Minute
ESP Electro Submersible Pump(s)
FER Fracture Extension Rate
HRWP High Rate Water Pack
PI Productivity Index
P_{net} Net Pressure

PPA	Pounds of Proppant Added per gallon of fluid
PSD	Particle Size Distribution
SDT	Step Down Test
SRT	Step Rate Test
SSP	Self-Supporting Proppant
TD	Total Depth
TSO	Tip Screen Out
UCS	Unconfined Compressive Strength

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